CHAPTER 10

Unit-Testing JavaScript

Unit testing for JavaScript is a big, somewhat contentious subject. For years, JavaScript's testing tools were significantly less powerful than Ruby's, and although I still think the Ruby tools are more flexible, the JavaScript tools are more than powerful enough to be useful.

And yet, many projects don't unit-test JavaScript code much, as there are significant impediments to writing good JavaScript tests:

- The ecosystem is still complicated. There are a lot of different tools, it's hard to differentiate between them, and setup is—as you'll see in a moment—still kind of challenging.
- The major frameworks don't have the same level of testing support that Rails does. To some extent this is getting better.
- A lot of JavaScript code is written to be strongly entangled with the DOM, making it harder to test the logic.

The last point is probably the most important, so let me elaborate on it.

In the discussion of Rails models and controllers, I talked about separating business logic from the database and from the Rails framework to make it easier to test the logic in isolation. I've also made the claim that isolating the business logic is good not just for testing, but also for making the long-term cost of change lower in the codebase.

Similarly, most JavaScript code ultimately is about responding to user action and changing elements in the DOM. And a lot of idiomatic JavaScript code completely intertwines business logic with the DOM in a way that makes the code hard to understand and change. Although I won't be covering the popular JavaScript frameworks here, they tend to do a good job of isolating business